

JOSHUA BAKATA

Mobile: +44 7301 586 778 Email: joshuabakata@gmail.com

LinkedIn: linkedin.com/in/joshuabakata

EDUCATION

Loughborough University, BSc (Hons) Computer Science & AI

2024–2027

First-year average: **82.3%**

modules include: Web Programming (React/JS, 90%), Software Engineering (89%), Formal Languages & Theory of Automata (89%), Embedded Systems (C++, 88%), Mathematics for Computer Science (87%).

Buckswood School, Hastings, East Sussex

2022–2024

A Levels: Mathematics (A*), French (A), Business (B)

RESEARCH

Predicting Power-Grid Outages with Deep Learning

Ongoing

Supervised by Dr Shirin Dora

- Using deep learning to predict power-grid outages, enabling operators to take preemptive action such as switching to backup systems, redirecting load, or preparing for downtime.
- Applying LSTMs and RNNs, which are well-suited to the long, noisy temporal patterns exhibited by grid signals.
- Exploring how the framing of the prediction task itself shapes the model's ability to learn the underlying dynamics.

PROJECTS

Offered Platform – Full-Stack Internship Project (Supabase, Flask, React, TailwindCSS) Jun. 2025 – Aug. 2025

- Built a student-facing SaaS to track internships, store applications, and manage personalised dashboards.
- Used Supabase (PostgreSQL) for authentication, row-level security, and real-time data access.
- Implemented a responsive React + Tailwind frontend with clean component architecture.
- Developed Flask microservices for secure API routing, session handling, and asynchronous tasks.
- Deployed the platform end-to-end and onboarded early users while iterating on product feedback.

Neural Network From Scratch (Python, NumPy)

Oct. 2025 – Dec. 2025

- Implemented a fully manual feed-forward neural network (no ML libraries) to classify Fashion-MNIST items.
- Coded backpropagation, gradient descent, and activation functions from first principles.
- Designed experiments to evaluate training stability, learning rates, and model accuracy.
- Used the project to deepen understanding of optimisation and ML fundamentals.
- Achieved a 96.85% test accuracy on unseen data.

HACKATHONS & TECHNICAL ENGAGEMENTS

- **HackNotts 2025 – LyricForge:** Built an AI remix agent that rewrote song lyrics to a chosen theme using Gemini and regenerated vocals with ElevenLabs, aligned via DTW.
- **G-Research Coding Challenge 2025:** Modelled and forecasted time-series data using ARMA models in Python, applying statistical reasoning to real-world datasets.

EXPERIENCE

Software Engineering Intern

Bitmovin, Klagenfurt

Incoming, Summer 2026

Video infrastructure · Y Combinator S15

- Incoming SWE Intern at Bitmovin, working within video encoding and streaming infrastructure.

TECHNICAL SKILLS

- **Languages:** Python, Java, C/C++, JavaScript, HTML/CSS, SQL
- **Frameworks & Tools:** React, Flask, Node.js, TailwindCSS, Git/GitHub, Streamlit
- **Libraries:** pandas, NumPy, scikit-learn, Matplotlib, ipywidgets
- **Other:** REST APIs, LLM integration, Machine Learning, Full-Stack Development

NON-TECHNICAL SKILLS

- **Leadership:** Coordinated student events as Prefect and House Captain, managing teams and logistics.
- **Public Speaking:** Delivered structured arguments in debates, later mentoring peers in presentation technique.
- **Teamwork:** Collaborated with prefect teams and faculty to organise school-wide events efficiently.
- **Organisation:** Managed budgets and timelines for inter-house projects, ensuring smooth execution.

INTERESTS

- **Mathematics:** Enjoy solving analytical problems and exploring applied probability models.
- **History:** Studying the evolution of military strategy and its influence on social development.

LEADERSHIP & SERVICE

- Completed the Civic and Military Leadership Programme in Rwanda, strengthening discipline, initiative, and resilience under pressure.
- Led a 52-member platoon through field operations, improving coordination, morale, and operational flow.
- Organised a panel for General Mk Mubarak, Chief of Defence Staff.
- Longest-serving commander in the battalion, trusted with ongoing supervision and team development.
- Served as House Captain (Y12) and Student Prefect (Y13), mentoring peers and representing student interests across school initiatives.